UNIT - V

FINANCIAL ACCOUNTING AND ANALYSIS

Subject: Managerial Economics & Financial Analysis (23A52402a)

Syllabus at a glance

- Accounting: concepts and conventions, double-entry bookkeeping
- Journal, ledger, trial balance
- Final accounts: trading A/c, profit and loss A/c, balance sheet with simple adjustments
- Financial analysis: liquidity, activity, capital structure and profitability ratios

1. Introduction to Accounting

1.1 Meaning

Accounting is the art of **recording**, **classifying**, **summarising** and **interpreting** financial transactions in monetary terms and communicating the results to users. (AICPA)

- **Book-keeping** = recording and classifying transactions (clerical). **Accounting** = book-keeping + summarising + analysis + interpretation.

1.2 Objectives and functions

- Systematic **record keeping**; finding **profit or loss**; knowing the **financial position** (assets and liabilities).
- Providing information to **owners**, **investors**, **lenders**, **government**, **employees**, **managers**; legal compliance (tax, Companies Act).

1.3 Branches

- **Financial accounting** (external reporting), **Cost accounting** (cost control), **Management accounting** (decision support).

1.4 Accounting cycle



2. Accounting Concepts and Conventions

2.1 Concepts (basic assumptions)

Concept	Meaning
Business entity	Business is separate from its owner; owner's capital is a liability of the business
Money measurement	Only transactions expressed in money are recorded
Going concern	Business will continue for the foreseeable future
Cost concept	Assets recorded at the purchase (historical) cost
Dual aspect	Every transaction has two effects: Assets = Liabilities + Capital
Accounting period	Profit measured for a fixed period (usually 1 year)
Matching concept	Expenses of a period are matched with revenue of that period
Realisation	Revenue is recognised when goods are sold / service given, not when cash is received
Accrual	Income and expenses recorded when earned / incurred
Objective evidence	Entries supported by vouchers and documents

Accounting equation: Assets = Liabilities + Capital (Owner's equity)

2.2 Conventions

- **Consistency** : same accounting methods are used year after year.
- **Conservatism (prudence)**: provide for all expected losses, ignore expected profits.
- **Full disclosure** : all material information must be shown in the statements.
- **Materiality**: record items which are significant; ignore trivial ones.

2.3 GAAP and Ind AS

- **GAAP** = Generally Accepted Accounting Principles. India follows **Accounting Standards (AS)** and **Ind AS** (converged with IFRS) issued by ICAI.

3. Double Entry Book-keeping

3.1 Meaning

Every transaction affects **two accounts** : one is debited and the other credited with an **equal amount** . Total debits = total credits.

3.2 Types of accounts and rules

Account	Type	Debit	Credit
Personal	Persons / firms (debtors, creditors)	Receiver	Giver
Real	Assets (cash, building, goods)	What comes in	What goes out
Nominal	Expenses, losses, incomes, gains	All expenses and losses	All incomes and gains

- Modern approach: **Assets and Expenses** : increase = Debit; **Liabilities, Capital and Income** : increase = Credit.

Account Name

DEBIT (Dr)	CREDIT (Cr)
To (increase of asset / expense	By (increase of liability / income)

3.3 Advantages and limitations

- Complete record, checks arithmetical accuracy (trial balance), shows profit and position, reduces fraud.
- Limitations: does not detect errors of principle, omission or compensating errors.

4. Journal

4.1 Meaning

- **Journal** = the book of original (prime) entry where transactions are recorded in **chronological order** with narration. Format: Date | Particulars | L.F. | Debit | Credit.

4.2 Specimen problem

Pass journal entries: (1) Started business with cash Rs. 1,00,000. (2) Purchased goods for cash Rs. 20,000. (3) Sold goods on credit to Ravi Rs. 15,000. (4) Bought furniture for cash Rs. 10,000. (5) Paid rent Rs. 2,000. (6) Received cash from Ravi Rs. 15,000.

No.	Journal entry	Dr (Rs.)	Cr (Rs.)
1	Cash A/c Dr. / To Capital A/c	1,00,000	1,00,000
2	Purchases A/c Dr. / To Cash A/c	20,000	20,000
3	Ravi A/c Dr. / To Sales A/c	15,000	15,000
4	Furniture A/c Dr. / To Cash A/c	10,000	10,000
5	Rent A/c Dr. / To Cash A/c	2,000	2,000
6	Cash A/c Dr. / To Ravi A/c	15,000	15,000

4.3 Subsidiary books

- **Cash book** (cash and bank), **Purchases book**, **Sales book**, **Purchase returns** and **Sales returns** books, **Bills receivable / payable** books and **Journal proper**.

5. Ledger

5.1 Meaning

- **Ledger** = the principal book containing **all accounts**; entries are **posted** from the journal. Balance of each account is found (balancing).
- Posting: debit side "To (account)", credit side "By (account)". Balance c/d and b/d.

5.2 Cash account from the problem above

Dr. CASH A/c	Rs.	Cr.	Rs.
To Capital	1,00,000	By Purchases	20,000
To Ravi	15,000	By Furniture	10,000
		By Rent	2,000
		By Balance c/d	83,000
	1,15,000		**1,15,000**

- Ravi A/c: Dr. side Rs. 15,000 (sales); Cr. side Rs. 15,000 (cash received); **balance nil**.

5.3 Journal vs Ledger

Basis	Journal	Ledger
Nature	Book of original entry	Book of final entry

Order	Chronological	Classified account-wise
Entry	Journalising	Posting
Narration	Required	Not required

6. Trial Balance

6.1 Meaning

Trial balance is a **statement of all ledger balances (debit and credit)** on a given date, prepared to check the **arithmetical accuracy** of the books. Total of debits = total of credits.

- Debit balances: assets, expenses, losses, drawings, purchases, debtors. Credit balances: liabilities, capital, incomes, sales, creditors.

6.2 Errors it does not disclose

- Errors of **omission**, **commission** (wrong account), **principle** (capital treated as revenue), **compensating** errors, and errors of original entry.

6.3 Specimen trial balance (used for final accounts below)

Debit balances	Rs.	Credit balances	Rs.
Opening stock	20,000	Sales	2,80,000
Purchases	1,50,000	Creditors	30,000
Wages	10,000	Capital	70,000
Carriage inward	5,000		
Salaries	20,000		
Rent	12,000		
Office expenses	8,000		
Furniture	30,000		
Plant and machinery	60,000		
Debtors	40,000		
Cash in hand	10,000		
Drawings	15,000		
Total	**3,80,000**	**Total**	**3,80,000**

7. Final Accounts

7.1 Components

- **Trading account** (gross profit), **Profit and Loss account** (net profit), **Balance sheet** (financial position).

7.2 Trading account: format

- Debit side: **Opening stock, Purchases (less returns), Direct expenses** (wages, carriage inward, freight, fuel), Gross profit c/d.
- Credit side: **Sales (less returns), Closing stock**, Gross loss c/d.

Gross profit = Sales + Closing stock - (Opening stock + Purchases + Direct expenses)

7.3 Profit and Loss account

- Debit side: indirect expenses (salaries, rent, office, depreciation, interest, bad debts); credit side: gross profit, other incomes (interest received, discount received, commission).

Net profit = Gross profit + Other incomes - Indirect expenses

7.4 Balance sheet

- **Liabilities** (left): capital + net profit - drawings, long-term and current liabilities. **Assets** (right): fixed assets, investments, current assets (in order of permanence or liquidity).

7.5 Adjustments (simple)

Adjustment	Treatment
Closing stock	Credit side of Trading A/c; asset in Balance sheet
Outstanding expense	Add to the expense in P&L; show as liability
Prepaid expense	Deduct from the expense in P&L; show as asset
Accrued (outstanding) income	Add to income; show as asset
Income received in advance	Deduct from income; show as liability
Depreciation	Debit in P&L; deduct from the asset in Balance sheet
Bad debts / Provision for doubtful debts	Expense in P&L; deduct from debtors
Interest on capital / drawings	Capital +, Drawings - (interest on drawings is income)

7.6 Solved problem (from trial balance above)

Adjustments: (a) Closing stock Rs. 30,000; (b) Outstanding salaries Rs. 2,000; (c) Prepaid rent Rs. 1,000; (d) Depreciation on plant and machinery @ 10%.

Trading account for the year ended 31 March

Dr. Particulars	Rs.	Cr. Particulars	Rs.
To Opening stock	20,000	By Sales	2,80,000
To Purchases	1,50,000	By Closing stock	30,000
To Wages	10,000		
To Carriage inward	5,000		
To Gross profit c/d	1,25,000		
Total	**3,10,000**	**Total**	**3,10,000**

Profit and loss account

Dr. Particulars	Rs.	Cr. Particulars	Rs.
To Salaries 20,000 + 2,000	22,000	By Gross profit b/d	1,25,000
To Rent 12,000 - 1,000	11,000		
To Office expenses	8,000		
To Depreciation (plant 10%)	6,000		
To Net profit (to capital)	78,000		
Total	**1,25,000**	**Total**	**1,25,000**

Balance sheet as at 31 March

Liabilities	Rs.	Assets	Rs.
Capital	70,000	Plant and machinery 60,000 - 6,000	54,000
Add: Net profit	78,000	Furniture	30,000
Less: Drawings	(15,000)	Closing stock	30,000
Net capital	1,33,000	Debtors	40,000
Creditors	30,000	Cash in hand	10,000
Outstanding salaries	2,000	Prepaid rent	1,000
Total	**1,65,000**	**Total**	**1,65,000**

Check: both sides of the balance sheet tally at Rs. 1,65,000.

8. Financial Analysis (Ratio Analysis)

8.1 Meaning and objectives

- **Financial analysis** = interpreting financial statements to assess **liquidity, solvency, activity and profitability**.
- **Ratio** = relationship between two figures expressed as a number, percentage or proportion.

8.2 Types of analysis

- **Horizontal** (trend, year-to-year), **Vertical** (common size, % of base), **Comparative statements**, **Ratio analysis**, **Fund flow and cash flow analysis**.

8.3 Classification of ratios



8.4 Liquidity ratios (short-term solvency)

Current ratio = Current assets / Current liabilities (ideal 2 : 1)

Quick (acid-test) ratio = (Current assets - Inventory - Prepaid) / Current liabilities (ideal 1 : 1)

Absolute liquid ratio = (Cash + Marketable securities) / Current liabilities (ideal 0.5 : 1)

8.5 Activity (turnover) ratios

Inventory turnover = Cost of goods sold / Average inventory ; Days = 365 / turnover

Debtors turnover = Credit sales / Average debtors ; Collection period = 365 / turnover

Creditors turnover = Credit purchases / Average creditors ; Payment period = 365 / turnover

Total assets turnover = Sales / Total assets ; Fixed assets turnover = Sales / Fixed assets

- Higher turnover = efficient use of assets; longer collection period = inefficiency in credit control.

8.6 Capital structure (leverage / solvency) ratios

Debt-equity ratio = Long-term debt / Shareholders' funds (ideal 2 : 1 or less)

Proprietary ratio = Shareholders' funds / Total assets

Interest coverage ratio = EBIT / Interest (higher is safer)

Capital gearing ratio = Fixed-interest capital / Equity capital

8.7 Profitability ratios

Gross profit ratio = (Gross profit / Net sales) x 100

Net profit ratio = (Net profit / Net sales) x 100 ; Operating ratio = (COGS + operating expenses) / Sales

Return on capital employed (ROCE) = (EBIT / Capital employed) x 100

Return on equity (ROE) = (Net profit after tax / Shareholders' equity) x 100

Earnings per share (EPS) = (Net profit - Preference dividend) / No. of equity shares

Price-earning ratio = Market price per share / EPS ; Dividend payout = DPS / EPS

9. Solved Ratio Problem

Data: Current assets Rs. 4,00,000 (inventory 1,00,000; cash 50,000; debtors 1,50,000); current liabilities Rs. 2,00,000 (creditors 90,000); sales (all credit) 12,00,000; COGS 9,00,000; gross profit 3,00,000; EBIT 1,80,000; interest 30,000; net profit 1,20,000; equity 5,00,000; long-term debt 3,00,000; total assets 10,00,000.

Ratio	Calculation	Result
Current ratio	4,00,000 / 2,00,000	2 : 1
Quick ratio	(4,00,000 - 1,00,000) / 2,00,000	1.5 : 1
Absolute liquid ratio	50,000 / 2,00,000	0.25 : 1
Inventory turnover	9,00,000 / 1,00,000	9 times (41 days)
Debtors turnover	12,00,000 / 1,50,000	8 times (46 days)
Creditors turnover	9,00,000 / 90,000	10 times (37 days)
Debt-equity ratio	3,00,000 / 5,00,000	0.6 : 1
Proprietary ratio	5,00,000 / 10,00,000	0.5 (50%)
Interest coverage	1,80,000 / 30,000	6 times
Gross profit ratio	3,00,000 / 12,00,000 x 100	25%
Net profit ratio	1,20,000 / 12,00,000 x 100	10%
ROCE	1,80,000 / (5,00,000 + 3,00,000) x 100	22.5%
ROE	1,20,000 / 5,00,000 x 100	24%
Total assets turnover	12,00,000 / 10,00,000	1.2 times

Interpretation

- **Liquidity**: current ratio 2 : 1 meets the standard; quick ratio 1.5 is comfortable, so the firm can pay short-term debts easily.
- **Activity**: stock moves every 41 days and debtors pay in 46 days; the firm pays creditors in 37 days, so it collects slower than it pays (needs credit control).

- **Capital structure**: low debt-equity (0.6) and interest cover of 6 times means low financial risk.
- **Profitability**: 25% gross margin and 10% net margin with ROE of 24% show good returns.

10. More Accounting Concepts and Problems

10.1 Capital vs Revenue items

Basis	Capital	Revenue
Nature	Long-term benefit (many years)	Benefit for the current year only
Examples	Purchase of machinery, land, building	Wages, rent, repairs, salaries
Shown in	Balance sheet	Trading and P&L account

- Wrong classification gives **errors of principle**, so profit is wrongly stated.

10.2 Depreciation

- **Depreciation** = decrease in the value of a fixed asset due to use, time and obsolescence.

Method	Formula
Straight line (SLM)	(Cost - Scrap value) / Life; equal charge every year
Written down value (WDV)	Fixed % on the book value at the start of each year

SLM: Cost Rs. 1,00,000; scrap Rs. 10,000; life 5 years: depreciation = 90,000 / 5 = **Rs. 18,000 per year**.
WDV @ 10%: Year 1 = 10,000 (WDV 90,000); Year 2 = 9,000 (WDV 81,000).

10.3 Bad debts and provision for doubtful debts

Debtors Rs. 41,000; bad debts Rs. 1,000 (to be written off); provision 5% on the remaining debtors.

- Bad debts = Rs. 1,000 (debit P&L). Provision = (41,000 - 1,000) x 5% = **Rs. 2,000** (debit P&L).
- Balance sheet: Debtors 41,000 - 1,000 = 40,000, less provision 2,000 = **Rs. 38,000**.

10.4 Cash basis vs Accrual basis

- **Cash basis**: record income and expense when cash is received or paid. **Accrual basis**:

record when earned or incurred (required by the Companies Act). Accrual shows true profit.

10.5 Problem: preparing a trial balance

Prepare a trial balance from these balances: Capital 50,000; Cash 5,000; Bank 30,000; Debtors 15,000; Creditors 12,000; Furniture 4,000; Purchases 60,000; Sales 80,000; Expenses 28,000.

Debit balances	Rs.	Credit balances	Rs.
Cash	5,000	Capital	50,000
Bank	30,000	Creditors	12,000
Debtors	15,000	Sales	80,000
Furniture	4,000		
Purchases	60,000		
Expenses	28,000		
Total	**1,42,000**	**Total**	**1,42,000**

- Gross profit (ignoring stock) = Sales 80,000 - Purchases 60,000 = 20,000; Net profit = 20,000 - 28,000 = **loss of Rs. 8,000**.

10.6 Common-size and comparative statements

- **Common-size statement**: every item shown as a % of a base (sales for P&L; total assets for balance sheet). **Comparative statement**: figures of two or more years side by side with absolute and % change.

Item	2023	2024	Change	% change
Sales	10,00,000	12,00,000	2,00,000	20%
Net profit	80,000	1,20,000	40,000	50%

- Profit has grown faster than sales, so **margins improved** (8% to 10%).

11. More Ratio Problems

Problem 1: Missing figures

Current liabilities Rs. 60,000; current ratio 2.5 : 1; quick ratio 1.5 : 1. Find current assets, quick assets and inventory.

- Current assets = 60,000 x 2.5 = **1,50,000** ; quick assets = 60,000 x 1.5 = **90,000** .

Inventory = 1,50,000 - 90,000 = Rs. 60,000

Problem 2: EPS and P/E

Net profit after tax Rs. 6,00,000; preference dividend Rs. 1,00,000; equity shares 50,000; market price Rs. 120; dividend per share Rs. 4.

EPS = (6,00,000 - 1,00,000) / 50,000 = Rs. 10

P/E ratio = 120 / 10 = 12 times; Dividend payout = 4 / 10 = 40%

Problem 3: Stock turnover

Opening stock Rs. 40,000; closing stock Rs. 60,000; sales Rs. 4,00,000; gross profit 25% of sales.

- Gross profit = 1,00,000; COGS = 4,00,000 - 1,00,000 = 3,00,000; average stock = (40,000 + 60,000)/2 = 50,000.

Stock turnover = 3,00,000 / 50,000 = 6 times; holding period = 365 / 6 = about 61 days

Problem 4: Capital structure

Equity share capital 4,00,000; reserves 1,00,000; 10% debentures 2,50,000; total assets 9,00,000. Find the debt-equity and proprietary ratios.

Shareholders' funds = 4,00,000 + 1,00,000 = 5,00,000; Debt-equity = 2,50,000 / 5,00,000 = 0.5 : 1

Proprietary ratio = 5,00,000 / 9,00,000 = 0.56 (56%)

Problem 5: Interpreting a ratio

Current ratio is 3 : 1 but quick ratio is 0.8 : 1. What does it show?

- Liquidity looks strong on the surface but **most of the current assets are tied up in inventory** ; the firm may struggle to pay immediate liabilities, so stock management needs attention.

12. Limitations of Ratio Analysis and Revision

- Based on historical data; affected by inflation and window dressing; no uniform definitions; ignores qualitative factors; needs comparison with industry norms and past trends.

Group	Main ratios	Shows

Liquidity	Current, quick, absolute	Short-term solvency
Activity	Inventory, debtors, creditors, assets turnover	Efficiency
Capital structure	Debt-equity, proprietary, interest cover	Long-term solvency
Profitability	GP, NP, ROCE, ROE, EPS	Earning power

Golden rules of accounts: Personal - debit the receiver, credit the giver. Real - debit what comes in, credit what goes out. Nominal - debit expenses and losses, credit incomes and gains.

13. Important Exam Questions

- Explain the accounting concepts and conventions. (10 M)
- What is double-entry book-keeping? State its advantages and rules for journal entries. (5 M)
- Journalise given transactions, post them into ledger and prepare a trial balance. (10 M)
- Prepare trading account, profit and loss account and balance sheet with adjustments. (10 M)
- Explain liquidity, activity, capital structure and profitability ratios with formulas. (10 M)
- Calculate ratios from the given balance sheet and interpret them. (10 M)